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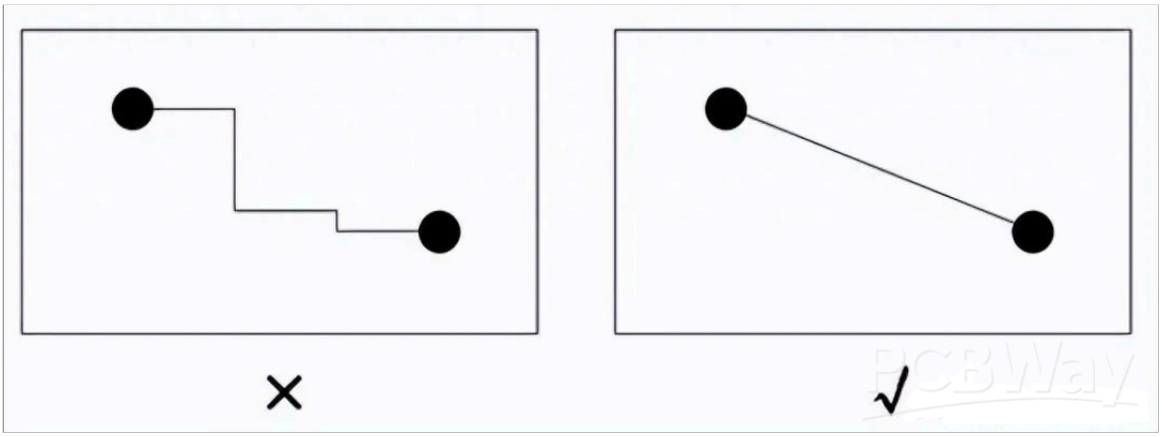
Common Rules for PCB Routing

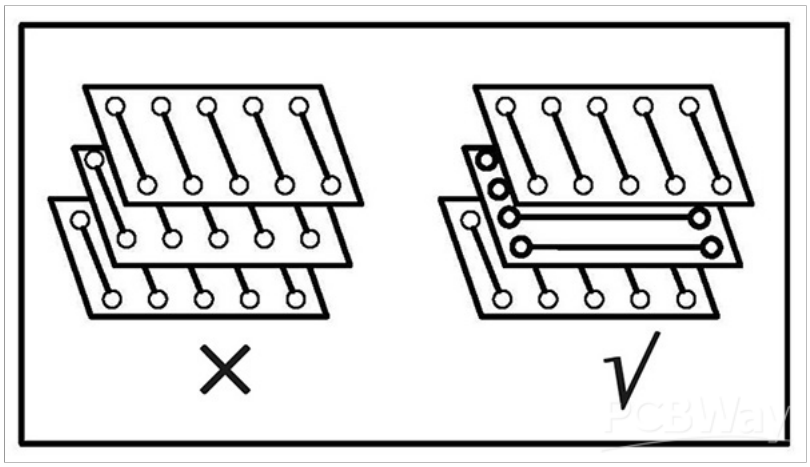
1. Length Control in Routing

Length controlling is very important in PCB design. The traces should be kept as short as possible to avoid introducing unnecessary interference due to excessive length. For important signal traces, such as clock signal traces, be sure to place its oscillator very close to the device.



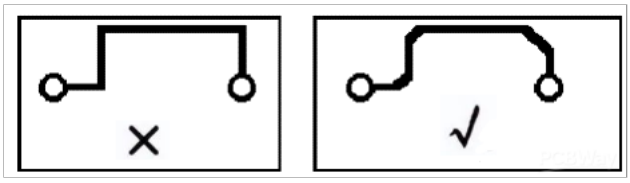
2. Direction Control in Routing

The routing directions of adjacent layers should be orthogonal to avoid the signal traces of adjacent layers running in the same direction, thus reducing unnecessary crosstalk between layers.



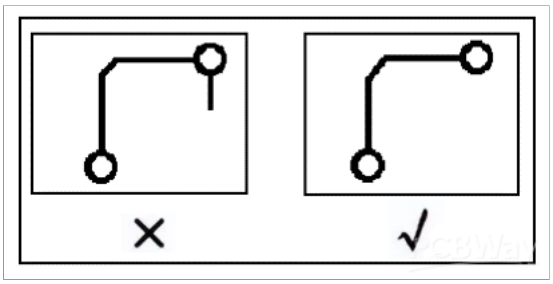
3. Rules for Chamfer

Sharp angles and right angles should be avoided in PCB design, otherwise, unnecessary radiation will be generated and the process performance will be poor.



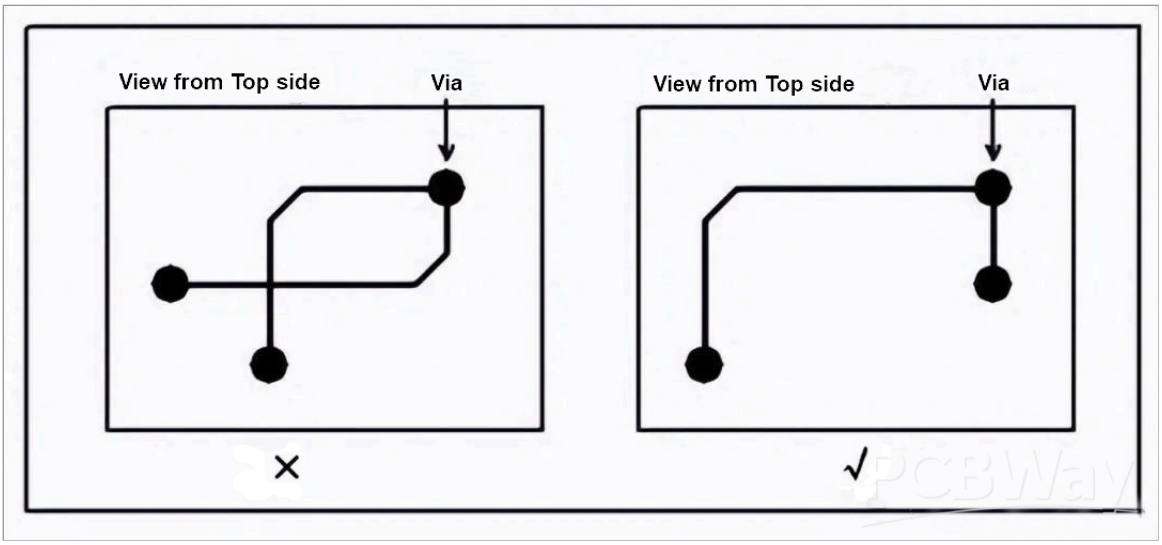
4. Open Loop Detection in Routing

Generally, dangling lines are not allowed. This is mainly to avoid the "antenna effect" and reduce unnecessary interference radiation and reception. Otherwise, serious consequences may occur.



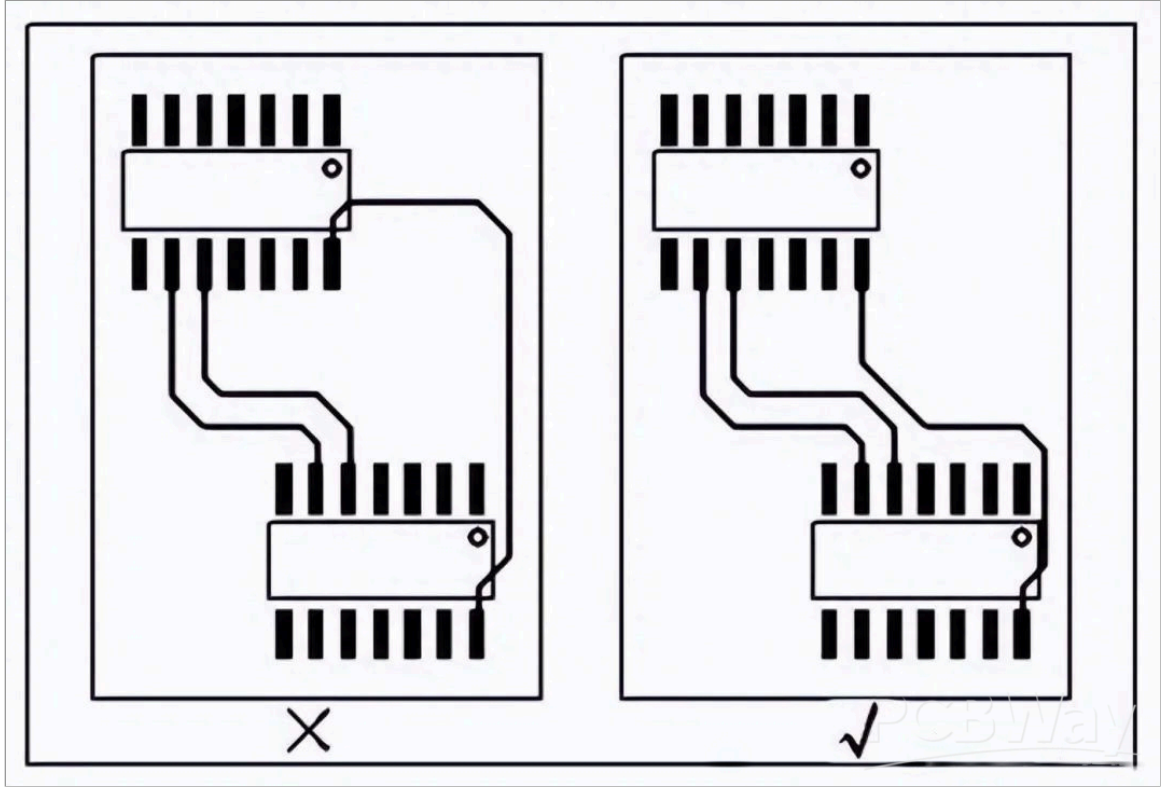
5. Try to Avoid Self-loop Routing

When designing PCB, we should pay attention to the situation where signal traces form self-loops between different layers. Especially when routing multi-layer boards, signal traces are routed between layers, which has a greater chance of forming self-loops. Self-loops can cause radiation interference.



6. Rules for Ground Loop

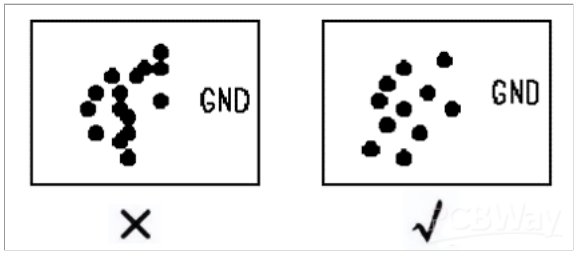
The minimal loop rule states that the loop area formed by a signal trace and its return path should be as small as possible. The smaller the loop area, the less external radiation it emits and the less interference it receives from the outside.



7. Rules for the Integrity of Power and Ground Layers

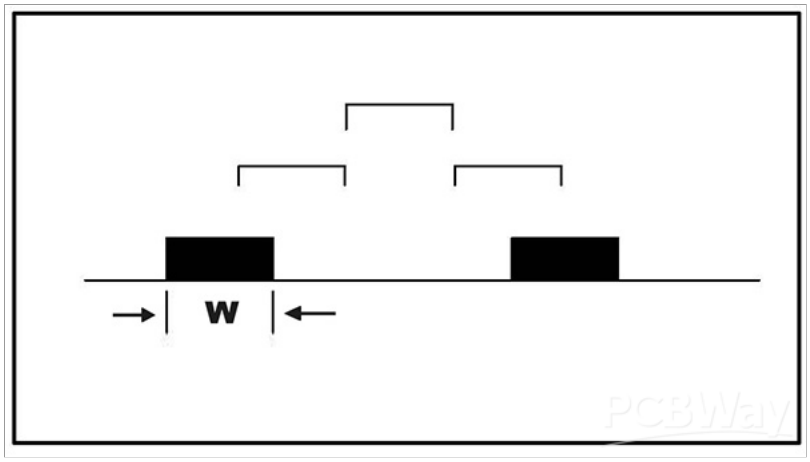
For areas with dense vias, care should be taken to avoid the holes connecting to each other in the hollowed-out areas of the power and ground layers, which would segment the plane layer.

This may destroy the integrity of the plane layer and further increase the loop area of the signal trace in the ground layer.



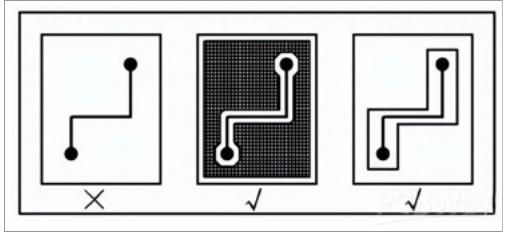
8. 3W Rule

In order to reduce crosstalk between traces, the trace spacing should be large enough. When the center-to-center distance between traces is at least three times the space width, 70% of the electric field will not interfere with each other, known as the 3W rule. To achieve 98% of the electric field not interfering with each other, a 10W spacing can be used.



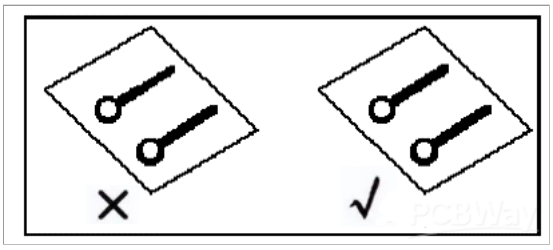
9. Shielding Protection

Shielding protection corresponds to the ground loop rule, which is actually to minimize the loop area of the signal. It is more common in some important signals, such as clock signals and synchronization signals. For some particularly important, high-frequency signals, a coaxial cable shielding design should be considered. This involves isolating the signal traces with ground traces on all sides (top, bottom, left, and right). Additionally, careful consideration must be given to how effectively to connect the shield ground to the actual ground plane.



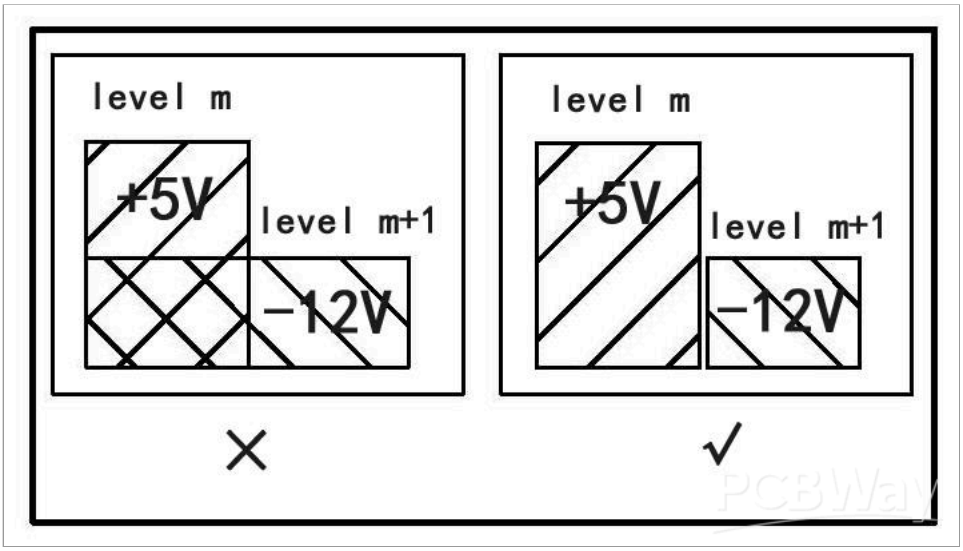
10. Rules for Impedance Matching Check

The routing width in the same network should be kept consistent. Changes in trace width can cause non-uniform characteristic impedance of the circuit, leading to reflections when the transmission speed is high. This should be avoided in the design whenever possible. Under certain conditions, such as structures similar to lead-out traces of connectors or the lead-out traces of BGA packages, changes in trace width may be unavoidable, so the effective length of the inconsistent sections should be minimized.



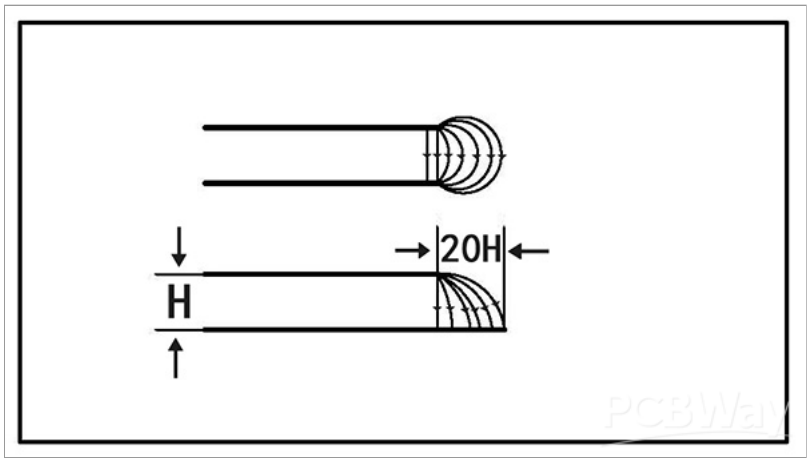
11. Rules for Overlapping Power and Ground Layers

Different power layers should avoid overlapping in space to reduce interference between different power supplies. This is especially important for power supplies with significantly different voltages. If it is difficult to avoid, consider placing a ground plane in between.



12. 20H Principle

To prevent board edge radiation, the power plane should be retracted during design. Using H (the dielectric thickness between the power and ground) as a unit, a retraction of 20H can confine 70% of the electric field within the edge of the ground layer, while a retraction of 100H can confine 98% of the electric field within it.



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